HANYU BAI

+1 734 263 3532 hanyubai@umich.edu

EDUCATION

University of Michigan *Jan. 2025 – Aug. 2027 (Expected)*
Ph.D. in Electrical and Computer Engineering Ann Arbor, MI, United States

National University of Singapore *Aug. 2022 – Dec. 2024 (Transferred)*
Ph.D. in Mechanical Engineering Singapore

National University of Singapore *Aug. 2021 – May 2022*
M.Sc. in Mechanical Engineering Singapore

University of British Columbia *Aug. 2016 – May 2021*
B.A.Sc. in Mechanical Engineering, With Distinction Vancouver, BC, Canada

EXPERIENCE

Battery R&D Intern *May 2026 – Aug. 2026*
Farasis Energy USA

- Develop online parameter-identification and state-of-health (SOH) diagnostic methods for EV lithium-ion batteries by integrating electrochemical models with lightweight machine-learning algorithms and routinely available BMS data, enabling deployment in cloud-based BMS systems and EV testing centers.
- Analyze the effects of driving behavior, duty-cycle variability, and climate on battery aging trajectories and model-parameter evolution; design targeted degradation-characterization experiments to validate diagnostic models and interpret parameter changes in relation to underlying aging mechanisms.

Graduate Research Assistant *Sep. 2022 – Present*
Energy Storage Lab; Advisor: Prof. Ziyou Song

- Develop physics-informed and machine-learning models to predict remaining lifetime of both fresh and aged lithium-ion batteries, supporting lifecycle-aware battery management and retirement, maintenance, and reuse decisions.
- Build battery lifecycle digital twins for multi-scenario simulations to model battery use from EV service through retirement and second-life applications, evaluating lifetime, techno-economic, and environmental outcomes to identify sustainable deployment and planning strategies.
- Develop AI-for-Science prediction frameworks that integrate physical knowledge with data-efficient learning and calibrated uncertainty quantification, providing reliable predictions and actionable confidence guidance for physical-system modeling.

Graduate Teaching Assistant *Aug. 2022 – Dec. 2024*
Department of Mechanical Engineering, National University of Singapore

- Led graduate laboratory sessions for ME5418 Machine Learning in Robotics, introducing reinforcement-learning fundamentals and supporting hands-on implementation and debugging of maze-based optimal path-planning algorithms.
- Facilitated undergraduate laboratories and tutorials for ME2142 Feedback Control Systems and ME2112 Strength of Materials, covering dynamic-system modeling, PID-controller tuning, static equilibrium, stress and strain, deformation, and structural analysis.

Graduate Research Assistant *Aug. 2021 – May 2022*
NUS Mechanical Engineering; Advisor: Prof. Seeram Ramakrishna

- Developed and optimized polyvinylpyrrolidone-lithium chloride (PVP-LiCl) composite desiccants through gravimetric water-sorption and kinetic characterization, identifying a 66.7 wt% composition with 280% equilibrium sorption capacity at 75% RH and 25 °C.
- Designed and evaluated PVP-LiCl-coated heat-exchanger prototypes under dynamic airflow conditions; demonstrated eightfold higher moisture-removal capacity than a silica-gel control and modeled sorption kinetics using the Elovich framework.

R&D Co-op *May 2019 – Sep. 2019*
Lock-Block Ltd.
- Conducted concrete mix-design trials and developed AutoCAD assembly models with 3D-printed prototypes to support Lock-Block design evaluation and iteration.
- Assessed the seismic performance of a 6-meter Lock-Block arch using ANSYS simulations, identifying the maximum ground acceleration before structural collapse.

R&D Co-op *Sep. 2018 – May 2019*
Ballard Power Systems
- Characterized gas-diffusion-layer (GDL) materials, fuel-cell diagnostic, and accelerated-stress tests to evaluate fuel-cell performance and durability for material selection.
- Analyzed test data to streamline diagnostic protocols and reduce testing time; authored facility procedures and user manuals and supported laboratory-equipment safety improvements.

ENTREPRENEURIAL EXPERIENCE

Co-Founder & Product Development Lead *May 2022 – Sep. 2023*
CARDY CRAFT
- Co-founded a sustainable pet-furniture venture developing modular, reconfigurable cat-climbing systems made from recyclable cardboard-based components.
- Designed the modular architecture and an interactive 3D configuration workflow that enabled customers to customize layouts, assess structural feasibility and safety, and generate component counts for purchasing.

Co-Founder & Technical Lead *Sep. 2021 – Aug. 2022*
Bikeye
- Co-founded Bikeye, a cycling-safety venture developing an mmWave-radar collision-warning system that detected nearby vehicles, estimated trajectories relative to the cyclist's projected path, and generated early avoidance alerts.
- Contributed to sensing architecture, trajectory-prediction logic, prototype development, and user-centered warning design.

ADDITIONAL EXPERIENCE

Pastry Chef *May 2019 – Sep. 2019*
Miku Vancouver, Michelin Guide Recommended
- Prepared and plated desserts for evening service, ensuring consistent presentation, quality, and timely delivery.

HONORS & AWARDS

- Finalist, College of Engineering Graduate Marian Sarah Parker Prize, University of Michigan (2026).
- Excellence in ECE Honor Roll, University of Michigan (2024–2025; 2025–2026).
- Third Prize, CDE Culinary Competition, National University of Singapore (2024).
- Dean's Honour List, University of British Columbia (2017, 2019).
- Outstanding International Student Award, University of British Columbia (2016).

PUBLICATIONS

- Zhang, J., Zhang, Y., Yi, B., Ren, Y., Jiao, Q., **Bai, H.**, Jiang, W., & Song, Z. "Discovery Learning predicts battery cycle life from minimal experiments." *Nature* **650**, 110–115 (2026). Link
- **Bai, H.**, Hu, Q., Ren, Y., Jiang, W., & Song, Z. "Battery Electrode-Level Diagnostics: Enabled by a 2-Minute Transient Response." *2025 IEEE Conference on Control Technology and Applications (CCTA)*, 97–102 (2025). Link
- Cheng, Y., **Bai, H.**, Liang, Y., Cui, X., Jiang, W., & Song, Z. "Data-Driven Quantification of Battery Degradation Modes via Critical Features from Charging." *arXiv:2412.10044* (2024). Link
- **Bai, H.**, Hu, X., & Song, Z. "The Primary Obstacle to Unlocking Large-Scale Battery Digital Twins." *Joule* **7**, 855–857 (2023). Link
- **Bai, H.**, & Song, Z. "Lithium-Ion Battery, Sodium-Ion Battery, or Redox-Flow Battery: A Comprehensive Comparison in Renewable Energy Systems." *Journal of Power Sources* **580**, 233426 (2023). Link
- **Bai, H.**, Lei, S., Geng, S., Hu, X., Li, Z., & Song, Z. "Techno-Economic Assessment of Isolated Micro-Grids with Second-Life Batteries: A Reliability-Oriented Iterative Design Framework." *Applied Energy* **364**, 123068 (2024). Link
- **Bai, H.**, Wee, M. G. V., Chinnappan, A., Li, J., Shang, R., & Ramakrishna, S. "Effect of Polyvinylpyrrolidone and Lithium Chloride Composite Desiccant-Coated Heat Exchangers on Dehumidification Studies." *Applied Thermal Engineering*, 123318 (2024). Link
- Gu, X., **Bai, H.**, Cui, X., Zhu, J., Zhuang, W., Li, Z., Hu, X., & Song, Z. "Challenges and Opportunities for Second-Life Batteries: Key Technologies and Economy." *Renewable & Sustainable Energy Reviews* **192**, 114191 (2024). Link
- **Bai, H.** "A Water-Saving Dust-Suppressing Sprayer Design for a Conveyor Belt." Chinese Patent CN 203997968 U (2014).

EXTRACURRICULAR ACTIVITIES

- **Student Leader, Academic Careers in ECE, University of Michigan:** Organize academic-career and professional-development activities for ECE graduate students.
- **Graduate Student Council Member, University of Michigan:** Support graduate-student community programming.
- **Authorized Signatory & Leadership Team Member, University of Michigan Chinese Women's Basketball Team:** Support organization governance, team operations, and event coordination as an authorized representative.
- **Teaching Assistant and Research Mentor, University of Michigan:** Delivered 300+ hours of instruction and mentored 11 undergraduate, graduate, and Ph.D. researchers.
- **Served as a peer reviewer for**: *Nature Communications*, *Nature Computational Science*, *Applied Energy*, *IEEE Transactions on Intelligent Transportation Systems*, *Green Energy and Intelligent Transportation*, *IEEE Transactions on Transportation Electrification*, *Automotive Innovation*, *IEEE Transactions on Power Electronics*, and *eTransportation*.

TECHNICAL SKILLS

Experimental and Engineering Tools: Electrochemical impedance spectroscopy (EIS), cyclic voltammetry (CV), fuel-cell assembly diagnostics, accelerated stress testing, dynamic vapor-sorption characterization, heat-exchanger testing, AutoCAD, ANSYS, 3D prototyping

Technical Modeling: Electrochemical and equivalent-circuit battery modeling, sorption-kinetics modeling, multi-objective optimization, techno-economic analysis

PERSONAL INTERESTS

- **Digital Illustration:** Independently create an original collection of digital stickers and meme illustrations, combining character design, humor, and visual storytelling.
- **Athletics:** Competed in a Beijing high-school girls' basketball league and placed eighth in the 400 m at the Xicheng District Primary and Secondary School Track and Field Meet (High School Division); practice Muay Thai (Level 1 Certificate), Brazilian Jiu-Jitsu, and skiing.
- **Music & Baking:** Play guzheng and electronic keyboard; previously operated an independent custom-dessert business producing made-to-order cakes and desserts, including options for customers with special dietary needs.